

Vithal Teja

Software Engineer

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Professional Summary

Software Engineer with 3+ years of hands-on experience building scalable cloud solutions at Amazon and other leading companies. Proven expertise in reducing system downtime by 65%, improving system scalability by 40%, and maintaining 99.9% uptime for high-traffic applications. Strong technical foundation in serverless systems using AWS Lambda, DynamoDB, and EventBridge; containerized applications with Kubernetes and Docker; and backend microservices development in Java, Spring Boot, and Python. Solid understanding of distributed systems architecture, high availability patterns, and multi-region deployment strategies. Proficient in designing and implementing CI/CD pipelines with Jenkins and GitHub Actions, Infrastructure-as-Code practices using Terraform and CloudFormation, and comprehensive monitoring solutions with CloudWatch. Experience with database optimization, query tuning, and caching strategies. Familiar with event-driven architecture, message queues (Kafka, SQS), and API Gateway integration. Passionate about learning cloud technologies, writing clean code, and collaborating with teams to deliver reliable systems. Eager to contribute to innovative backend and cloud infrastructure projects while continuously growing technical skills. Master's in Computer Science with focus on cloud systems design.

Technical Skills

- Programming Languages**: Java, Python, C++, Go.
- Web & Application Framework**: Spring Boot, Node.js, Django, React.js, Angular, HTML, JavaScript, TypeScript, CSS, Bootstrap.
- Data & Analytics Tools**: SQL, PostgreSQL, MySQL, MongoDB, Hadoop, DynamoDB, TensorFlow, Scikit-learn, Cognitive Services, Data Pipelines, Pandas, NumPy.
- Cloud Platforms DevOps**: Microsoft Azure (Functions, App Services, Blob Storage, Cosmos DB, Data Factory), AWS (EC2, S3, Lambda, RDS, DynamoDB), Docker, Kubernetes, Jenkins, Terraform, CloudFormation, Serverless Architecture, GCP.
- Monitoring & Automation**: Prometheus, Grafana, Ansible, Redis, Azure Monitor.
- Development Tools & Version Control**: Git, GitHub, GitHub Copilot, Jira, Agile (Scrum), TDD, CI/CD pipelines.
- Systems & Architecture**: Linux/Unix, Networking, Virtualization, Object-Oriented Design, Software Design, Software Development Life Cycle (SDLC), computer architecture, coding.

Professional Experience

- Amazon, System Development Engineer****Nashville, TN | July 2025 - Present**
- Architected serverless KPI platform on AWS CDK with Lambda, DynamoDB, and EventBridge, automatically processing and analyzing key performance indicators to drive data-driven business decisions, improving accuracy by 25% and reducing manual monitoring overhead by 40%.
 - Built real-time data pipeline using DynamoDB streams and Lambda functions to ingest, process, and intelligently categorize KPI scores with threshold-based filtering (Low/Medium/High), enabling continuous tracking of critical business metrics across the organization.
 - Designed comprehensive CloudWatch monitoring dashboards providing real-time visibility into KPI trends, score distributions, system health metrics (CPU, memory, disk utilization), and onboarding impact analytics, enabling proactive issue detection and rapid incident response.
 - Implemented EventBridge automation orchestrating continuous KPI updates and alert workflows, eliminating manual monitoring overhead and ensuring real-time insights are seamlessly distributed across all stakeholder teams.
 - Developed algorithm comparison framework (V1 vs V2 scoring methodologies) enabling rapid performance analysis, comparative evaluation, and system health assessment before production deployment, reducing risk during optimization changes.
 - Automated end-to-end CI/CD integration with Infrastructure-as-Code (AWS CDK), Git version control, and deployment automation, streamlining the release process, reducing deployment overhead, and enabling rapid iteration cycles.
 - Collaborated with cross-functional teams (Product, Analytics, Quality Assurance) delivering 100% of KPI Quality Manager requirements including trend analysis, intelligent filtering, and performance monitoring, achieving consistent on-time delivery across all sprint cycles.

- Geico, Java Full-Stack Developer****Atlanta, GA | Dec 2023 - June 2025**
- Built distributed Spring Boot microservice deployed across multiple AWS availability zones with high availability patterns and circuit breakers to monitor Kubernetes pods and AWS infrastructure, detecting health anomalies and triggering intelligent alerts and automated rollbacks reducing downtime by 65%.
 - Integrated with Kubernetes API to track pod health, restarts, crash loops, and resource usage enabling real-time visibility and observability across distributed containerized environments with multi-region monitoring capabilities.
 - Implemented secure IAM policies for Lambda, EKS, and CloudWatch enforcing least-privilege access and ensuring enterprise-grade security posture across distributed AWS services and infrastructure components.
 - Developed Java-based Lambda functions with message queue integration (AWS SNS/SQS) to automate anomaly detection, alert triggering, and rollback execution on service health breaches, preventing cascading failures in distributed microservices architecture with guaranteed message delivery.
 - Designed highly available, horizontally scalable architecture across multiple AWS regions and availability zones with Application Load Balancer (ALB), auto-scaling groups, Infrastructure-as-Code (Helm charts, CloudFormation), readiness/liveness probes, and failover patterns enabling zero-downtime deployments.
 - Exposed comprehensive RESTful APIs via AWS API Gateway with CloudFront CDN caching and global edge locations, displaying real-time uptime metrics, error rates, restart counts, and remediation suggestions for cross-functional team access with sub-100ms latency globally.
 - Automated comprehensive testing (JUnit, Mockito, REST-Assured) for functional, performance, and destructive failure scenarios validating distributed microservice resilience, high availability, and auto-recovery capabilities across multi-region deployments.

- HCL Tech, Software Engineer****Hyderabad, India | June 2021 - July 2022**
- Engineered highly available distributed RESTful microservices using Java, Spring Boot, and Python deployed behind AWS API Gateway and CloudFront CDN with Application Load Balancer for traffic distribution, processing 50,000+ requests/minute with 99.9% uptime across multiple availability zones and improving system throughput by 30%.
 - Designed cloud-native solutions on AWS (EC2, S3, RDS, Lambda) with Infrastructure-as-Code (Terraform, CloudFormation), Application Load Balancer (ALB), auto-scaling groups, multi-region failover patterns, and circuit breaker design patterns, increasing system reliability by 35% and reducing manual operations by 40%.
 - Built containerized microservices using Docker and Kubernetes with service mesh patterns, high availability configurations across multiple zones, and graceful degradation strategies, enhancing horizontal scalability by 40% and enabling independent service deployments with reduced conflicts and improved resilience.
 - Implemented CI/CD pipelines via Jenkins and GitHub Actions with Infrastructure-as-Code (Terraform, CloudFormation) for automated infrastructure provisioning, message queue integration (SQS, Kafka) for asynchronous deployments, comprehensive automated testing (JUnit, Mockito), and SonarQube integration, cutting release times by 30% and production bugs by 25%.
 - Optimized database performance through advanced PostgreSQL/MySQL query tuning, index optimization strategies, connection pooling, and distributed Redis caching, reducing database load by 45%, improving data retrieval speeds by 3x, and ensuring sub-100ms response times for 99th percentile queries in high-traffic scenarios.
 - Developed Python-based asynchronous processing services with Flask, Celery, and message queue integration (SQS, Kafka), handling 100,000+ daily background jobs with 99.5% success rate, implementing dead-letter queue (DLQ) patterns and retry logic for reliable distributed job processing.
 - Collaborated in Agile/Scrum sprints with cross-functional teams using Jira and Git, achieving 95% on-time delivery across 15+ sprints while maintaining high system reliability and uptime standards for distributed production systems.

Education

- Master of Science in Computer Science, GPA: 3.52/4.0****Kennesaw State University | Aug 2022 - Dec 2023**
- Bachelors in Computer Engineering, GPA:3.26/4.0****K L University, India | Aug 2017 - Mar 2021**

Certifications

- AWS Certified Developer – Associate
- Microsoft Certified: Azure Fundamentals (AZ-900)